Shambhawi Baswaraj Bhure

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Education

Columbia University — New York, NY
M.S. in Computer Science (ML Track), GPA: 4.04/4.00 — *Aug. 2025 – Dec. 2026*

Indian Institute of Technology, Guwahati — Assam, India
B.Tech in Electronics & Electrical Engineering, GPA: 8.36/10.00 — *Aug. 2019 – May 2023*

Experience

Tesla Inc. — Fremont, CA
Machine Learning Engineering Intern — *May 2026 – Aug. 2026*

- Engineered execution-verified Text-to-SQL/PromQL training pipelines, curating 1,900+ and 350+ gold examples; lifted SQL accuracy 4.7x and PromQL metric-match 88% to 98%.
- Architected a custom quantization and LoRA stack to fine-tune an unsupported mixture-of-experts model; shipped a self-correcting generation path with 38/40 production-baseline agreement.

Mastercard Technology Pvt. Ltd. — Pune, India
Software Engineer 2 — *Nov. 2024 – Jul. 2025*

- Hardened database security by migrating to SHA-256 encryption, cutting computational overhead by 4% across millions of daily fintech transactions.
- Led end-to-end design of *Ledger*, a distributed Java/Spring Boot microservice consolidating 8+ modules for Settlement Facilitator and Reconciliation teams.

Software Engineer 1 — *Jun. 2023 – Nov. 2024*

- Built Kafka-based Axon event adaptor for distributed event streaming, reducing codebase complexity by 50%; authored KT docs onboarding 5+ engineers.
- Shipped 5 features and resolved 10+ critical bugs for Settlement Facilitator with zero SLA breach; maintained 95%+ code coverage via JUnit and Mockito in CI/CD pipelines.

Software Development Intern — *May 2022 – Jul. 2022*

- Built real-time monitoring dashboard tracking 5+ application metrics using Prometheus & Grafana; extended Splunk dashboard with 10+ custom transaction-monitoring queries.
- Configured automated alerting for critical service thresholds, reducing mean time-to-detect (MTTD) incidents across production fintech services.

Projects & Research

Workflow Optimizations for Agentic Pipelines | *LangGraph, MCP, Python, IBM Research* — *Jan. 2026 – Present*

- Architected a temporal semantic caching layer for agentic plan-execute pipelines, resolving time/parameter-sensitive invalidation gaps in standard caching, achieving 30.6x median speedup on cache hits.
- Devised disk-backed tool-discovery caching with dependency-aware parallel execution across MCP workflows, cutting median end-to-end latency by 40% (1.67x speedup); co-authored findings arXiv:2605.20630.

Agentic Developer Copilot | *Next.js, FastAPI, LangGraph, RAG, Pinecone, Docker* — *Jan. 2026 – May 2026*

- Built a full-stack agentic AI app orchestrating a multi-agent Corrective RAG (CRAG) pipeline with LangGraph and Llama 3.1 on Groq, reducing hallucinations by 95% via autonomous web-search fallback.
- Automated vector ingestion using HuggingFace embeddings and Pinecone, enabling sub-second semantic search across 500+ chunked documents; cut developer doc-lookup time by 40%.

Face Liveness & Plagiarism Detection | *CNN, Bi-LSTM, TensorFlow, NLP* — *Feb. 2024 – Aug. 2025*

- Built CNN-based Face Liveness Detection model (96% accuracy) benchmarked across 5 Transfer Learning architectures: VGG16, VGG19, ResNet50, DenseNet121, InceptionV3.
- Trained Bi-LSTM Plagiarism Detection model (83% accuracy); submitted multimodal academic integrity findings to *IEEE MultiMedia*.

Technical Skills

Languages: Python, Java, C++, SQL (PostgreSQL, MySQL), JavaScript, TypeScript, PromQL
AI / ML: PyTorch, TensorFlow, LangGraph, LangChain, RAG, HuggingFace, Pinecone, OpenAI, CNN, NLP, fine-tuning
Frameworks: Spring Boot, vLLM, LoRA/QLoRA, FastAPI, Flask, Next.js, React, Node.js
Cloud & DevOps: AWS, GCP, Azure, Docker, Kubernetes, Kafka, Microservices, CI/CD
Tools: Git, Linux, Agile/Scrum, Confluence, Sonar

Leadership & Initiatives

Teaching Assistant, Analysis of Algorithms & Intro to Databases (Fall 26), ML for Data Science (Spring 26).
Claude Hackathon 2026, 2nd Place for developing a real-time multi-agent AI system for debating city legislation.
Executive Head, Entrepreneurship Club, IIT Guwahati – Led club operations and mentored 20+ student founders.
Volunteer, Girls4Tech & Youth4Development – Delivered STEM and financial literacy sessions to 100+ girls.